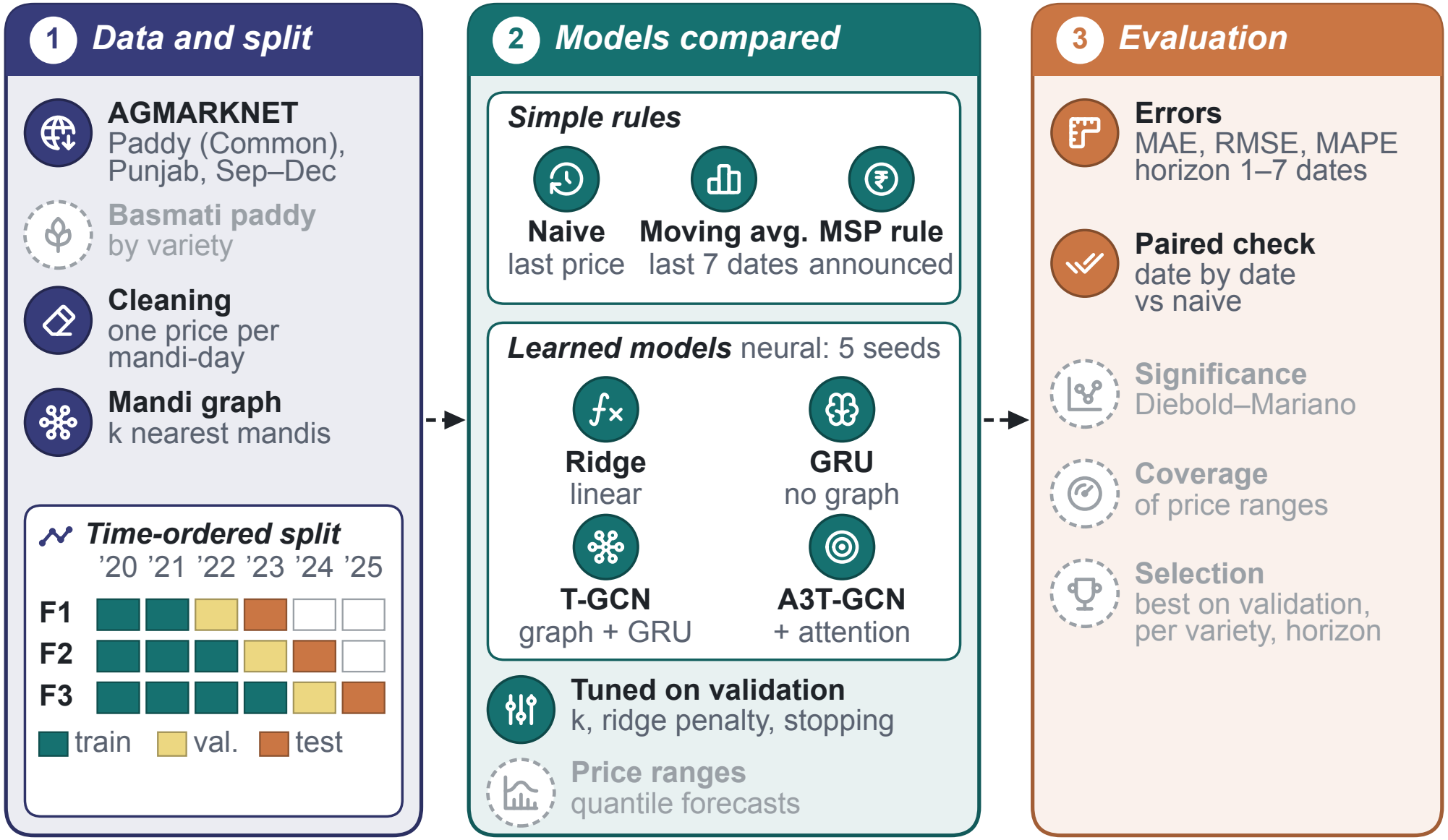

1 Data and split
AGMARKNET
Paddy (Common), Punjab, Sep–Dec
Basmati paddy
by variety
Cleaning
one price per mandi-day
Mandi graph
k nearest mandis
Time-ordered split
'20 '21 '22 '23 '24 '25
F1
F2
F3
train
val.
test
2 Models compared
Simple rules
Naive
last price
Moving avg.
last 7 dates
MSP rule
announced
Learned models neural: 5 seeds
Ridge
linear
GRU
no graph
T-GCN
graph + GRU
A3T-GCN
+ attention
Tuned on validation
k, ridge penalty, stopping
Price ranges
quantile forecasts
3 Evaluation
Errors
MAE, RMSE, MAPE
horizon 1–7 dates
Paired check
date by date
vs naive
Significance
Diebold–Mariano
Coverage
of price ranges
Selection
best on validation,
per variety, horizon